

Structure and language: First class

This is a well-organised report with a logical structure that facilitates readability throughout. The material is presented in a coherent manner, and the use of chapters, sections, and subsections is generally very effective. There is a clear logical progression through the report; for example, the individual techniques are introduced and developed in Section 5 before being combined and applied in Section 6. One exception is on page 21, where the introduction of sub-subsections (3.2.2.1 and 3.2.2.2) appears unnecessarily detailed and somewhat disrupts the otherwise clean structure of the report. The concluding chapter is largely devoted to a useful discussion of possible future work. However, the report would benefit from a more substantial summary of the work that has actually been carried out and a more explicit discussion of how the results address the research questions posed in the introduction. Figures 1 and 2 are well presented; however, the temporal aspect relevant for citation networks could have been introduced more explicitly. In Figure 4, the background shading should be removed, as it makes the figure appear unnecessarily cluttered. Figures 5 and 6 are particularly strong. The boxes in the upper-left panels should, however, be removed or repositioned since they obscure part of the data. If subplot captions are retained in Figure 4, it would be more informative for them to describe the relationship being illustrated rather than simply referring to equation numbers. The report demonstrates excellent spelling and grammar throughout and is clearly written carefully and independently by the student. The notation is largely consistent, although the alternating use of (x,y) and (a,b) for similar quantities is somewhat confusing for the reader. A simpler and more consistent notation would improve readability further. The reference list is well presented and generally consistent. A minor point is that complete consistency of bibliographic style (for example, in the use of author initials and first names) would improve the presentation further. Another minor stylistic issue concerns the punctuation of displayed equations. Equations should be punctuated according to the grammar of the surrounding sentence, and commas immediately preceding equations are occasionally used where they are not grammatically appropriate. Overall, the report is professionally presented, logically structured, and written to a very high standard.

Clarity: Upper second class

The report is somewhat mixed with respect to clarity. In several places the explanations are very strong; for example, the discussion surrounding Eq. (3) is particularly well executed, with the reader guided carefully through the meaning and implications of the equation. The introduction, layperson's summary, and parts of the results section are especially accessible and demonstrate a good awareness of the intended audience. In contrast, some later equations and concepts are not discussed with the same level of

care and accessibility. For example, Eq. (6) is introduced rather abruptly, with several symbols only defined afterwards. In addition, the report does not explicitly point out that the attachment probability is preferential through its proportionality to (k) , which is an important conceptual feature of the model. Technical terminology is generally explained when introduced, but in some cases simpler and more intuitive explanations would have improved accessibility for a non-specialist physics audience. More generally, the report would benefit from a clearer explanation of how the constructed model contributes to answering the research questions posed in the introduction. Two points that can ultimately be inferred from the report, but are not clearly explained at the outset, are: (i) that meaningful comparisons between systems of different sizes require a family of systems generated according to common underlying rules, and (ii) that running the model for (N) steps naturally generates systems of different sizes that can then be compared with theoretical predictions for infinite systems. There is also a mismatch between the introduction, where the research is very effectively motivated through citation networks and homophily associated with author characteristics, and the later chapters where these connections are only rarely revisited. The clarity of the report would have been improved by more frequent references back to this motivating application. For example, Figure 4 could have included discussion of the implications of the observed trends for citation networks, while the discussion of Figures 5 and 6 could have explored the practical significance of the agreement between simulations and theory in the context introduced earlier. As noted elsewhere, the alternating use of $((x,y))$, $((a,b))$, and later (x) with somewhat different meanings introduces unnecessary confusion. In particular, the variable (x) used in Section 4.3 and later sections is not explicitly defined, and a clearer treatment of notation would improve readability. On the positive side, the use of references is appropriate throughout, and the qualitative interpretation of the numerical results, particularly in Figures 5 and 6, is very good and demonstrates a solid understanding of the physical implications of the findings. Overall, while substantial parts of the report are clear and well explained, greater effort to maintain accessibility throughout and to connect the technical developments more explicitly to the motivating research questions would have strengthened the report considerably.

Techniques: First class

Overall, the report demonstrates a very good application of both theoretical and numerical techniques. The combination of analytical work with numerical simulations is handled effectively, and the comparison between theoretical predictions and numerical results is convincing. While not all theoretical derivations were straightforward to follow in complete detail, the overall methodology is sound and demonstrates strong technical competence. The integration of theory and simulation

works particularly well and strengthens the analysis considerably. There are, however, a few technical presentation issues. In Figure 4, it is not optimal to display the range $(0.01 \leq k \leq 1)$, since (k) represents an integer-valued quantity. In addition, it is unclear how fluctuations of order (10^{-12}) can arise in a system containing only (10^4) nodes, and this point would benefit from further explanation or clarification.

Scientific achievements: First class

The project demonstrates a strong scientific outcome overall. The student formulates a novel model incorporating cumulative advantage together with homophily associated with node attributes, and derives theoretical predictions for the asymptotic behaviour in the limit $(N \rightarrow \infty)$. These theoretical results are then compared systematically with numerical simulations of finite systems. The report also includes a review of goodness-of-fit procedures and proposes a modified goodness-of-fit approach tailored to the present context. This demonstrates a commendable level of independent scientific thinking. Overall, the project shows a strong grasp of the relevant scientific ideas and combines analytical and numerical work effectively. The achievements are scientifically very good and place the work at the boundary between a strong Upper Second Class and a First Class standard.

Layperson's summary: First class

Overall, this is a very good layperson's summary. It provides an accessible account of the work undertaken and clearly outlines the central research question addressed in the project. The summary succeeds in communicating the broader motivation and context of the work to a non-specialist audience. The summary is engagingly written and pitched at an appropriate level for the intended audience of A-level Physics students. It makes a clear effort to explain the significance of the project while maintaining accessibility, and it provides a coherent overview of the motivation, approach, and broader context of the work. The final three paragraphs contain several bracketed comments. It would be preferable to avoid excessive use of such comments, as they tend to read more like notes to the author than part of a polished summary intended for publication-style presentation. Where the comments are not essential, they should simply be omitted; where they are important, they should instead be incorporated naturally into the main text. Overall, the summary is well written, accessible, and engaging, although there remains some scope for improvement in terms of overall polish and presentation.